# Chapter 11
# Liquids & Intermolecular Forces

11.1 A Molecular Comparison of Gases, Liquids and solids

11.2 Intermolecular Forces, IMF

11.3 Select Properties of Liquids

11.4 Phase Changes

11.5 Vapor Pressure

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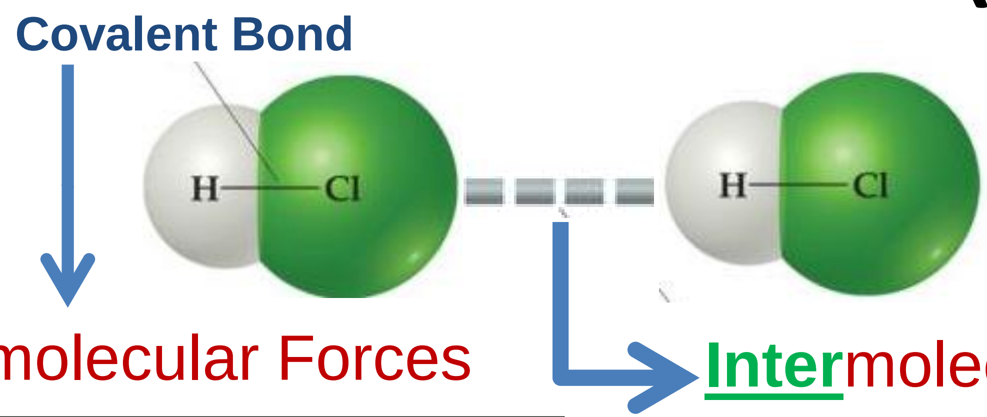
# Intermolecular Forces (IMF)




## Intramolecular Forces

- Forces that hold atoms together within a molecule (ionic, covalent & metallic bonds)
- Determines the bond strength.
- Generally stronger than (IMF)

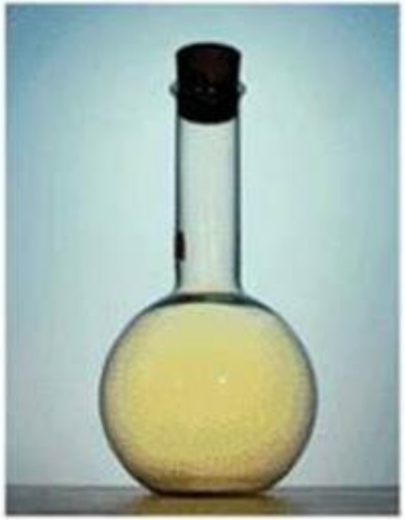
## Intermolecular Forces

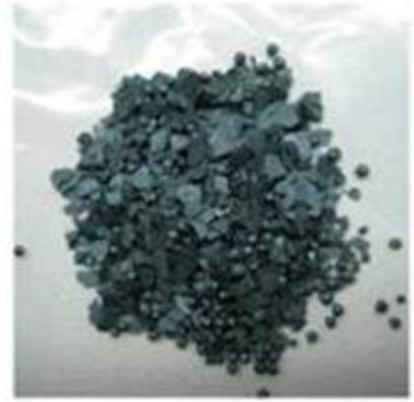
- Attractive forces between molecules.
- Determines the state of the matter.
- Responsible for melting & boiling points.

$HCl(g) \rightarrow H(g) + Cl(g)$ $\Delta H_{bond\ enthalpy} = 431\ kJ/mol$

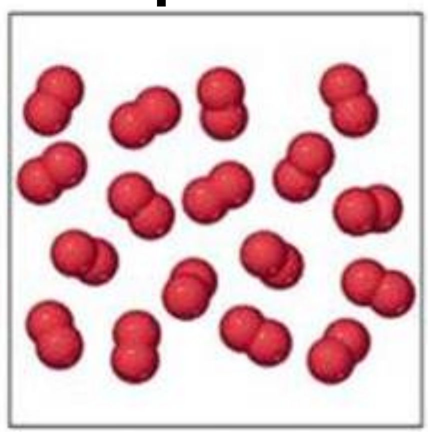
$HCl(L) \rightarrow HCl(g)$ $\Delta H_{vap} = 16\ kJ/mol$

# 11.1 A Molecular Comparison of Gases, Liquids and solids



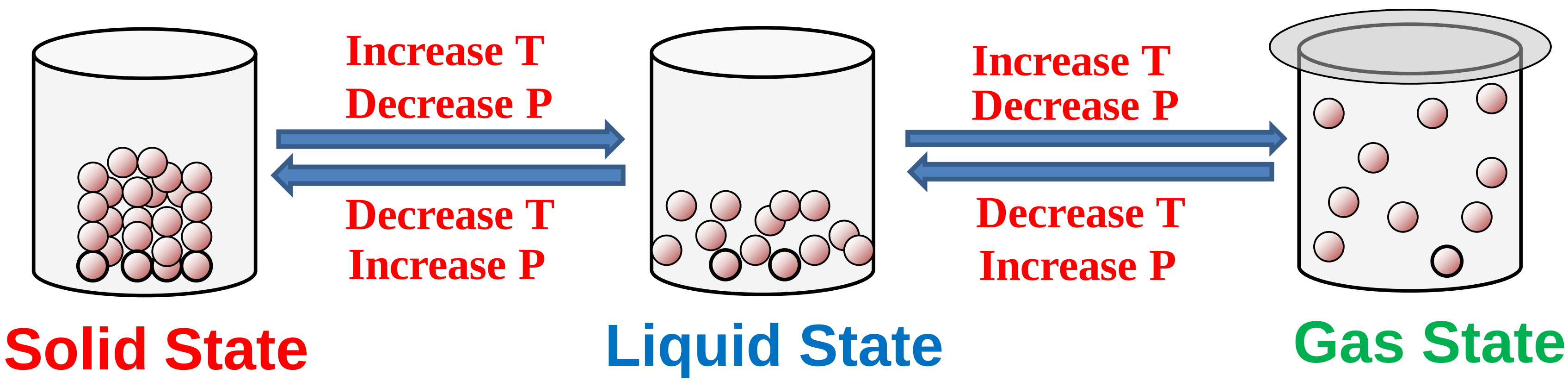


**Strength of intermolecular attraction increases**

| Gas | Liquid | Solid |
|---|---|---|
|  |  |  |
| $Cl_2$ | $Br_2$ | $I_2$ |

# 11.1 A Molecular Comparison of Gases, Liquids and solids

- Fundamental difference between states of matter is the strength of intermolecular forces attractions.
- Stronger forces bring molecules together.
- Kinetic energy keeps them apart and moving.




$HCl(L) \rightarrow HCl(g)$ $\quad \Delta H_{vap} = 16$ kJ/mol

$HCl(g) \rightarrow H(g) + Cl(g)$ $\quad \Delta H_{bond} = 431$ kJ/mol

# Some Characteristic Properties of the States of Matter

## Solid State

- Has a definite shape
- Has a definite volume.
- Incompressible.
- Does not flow.
- Diffusion within a solid occurs extremely slow.
- Very strong interactions between particles.

- Takes the shape & the volume of its container

## Liquid State

- Takes the shape of its container.
- Has a definite volume
- Incompressible.
- Flow readily.
- Diffusion within a liquid occurs slowly.
- Strong interactions between particles.

## Gas State

- Takes the shape of its container.
- Takes the volume of its container.
- Compressible.
- Flow readily.
- Diffusion within a gas occurs rapidly.
- Negligible interactions between particles.

# 11.2 Intermolecular Forces, IMF

- IMF much weaker than intramoleculare forces.

**$HCl(L) \rightarrow HCl(g)$ $\Delta H_{vap}$ = 16 kJ/mol**

**$HCl(g) \rightarrow H(g) + Cl(g)$ $\Delta H_{bond\ enthalpy}$ = 431 kJ/mol**

**TABLE 11.2 Melting and Boiling Points of Representative Substances**

| Force Holding Particles Together | Substance | Melting Point (K) | Boiling Point (K) |
|---|---|---|---|
| *Chemical bonds* | | | |
| Ionic bonds | Lithium fluoride (LiF) | 1118 | 1949 |
| Metallic bonds | Beryllium (Be) | 1560 | 2742 |
| Covalent bonds | Diamond (C) | 3800 | 4300 |
| *Intermolecular forces* | | | |
| Dispersion forces | Nitrogen ($N_2$) | 63 | 77 |
| Dipole–dipole interactions | Hydrogen chloride (HCl) | 158 | 188 |
| Hydrogen bonding | Hydrogen fluoride (HF) | 190 | 293 |

- The stronger the attractive forces, the higher the boiling points of liquids and the higher the melting points of solids.

# Types of IMF

1. **Van der Waals Forces**
   - **Dispersion (London Dispersion) Forces**
   - **Dipole-Dipole Forces**

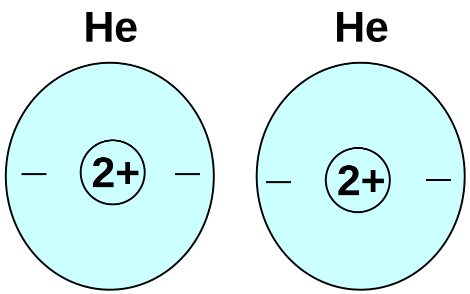
2. **Hydrogen Bonding Forces**

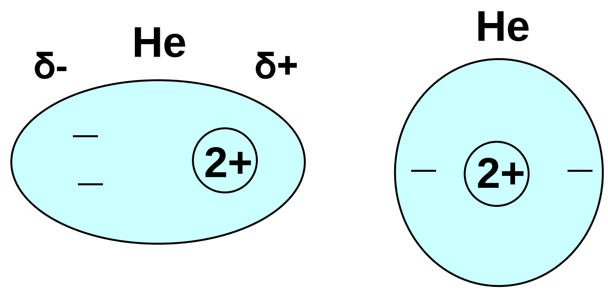
**(A special dipole-dipole forces)**

3. **Ion-Dipole Forces: important in solution**

# London dispersion forces

- **Attractive forces between all molecules**
- **Only forces between nonpolar covalent molecules**



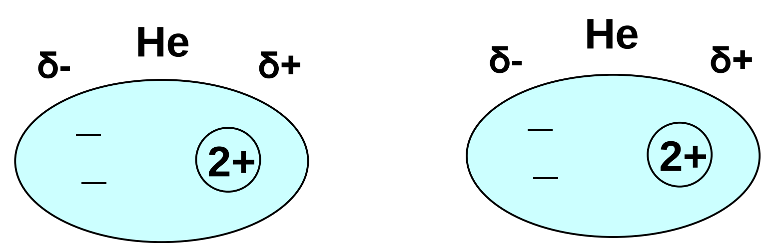




**spherical symmetrical electron distribution**

**instantaneous dipole moment**

**no Polarity**

**no permanent dipole**




**instantaneous dipole moment**

**induced instantaneous dipole moment**

**electrostatic attraction**

**Dispersion force**
**or**
**London dispersion force**

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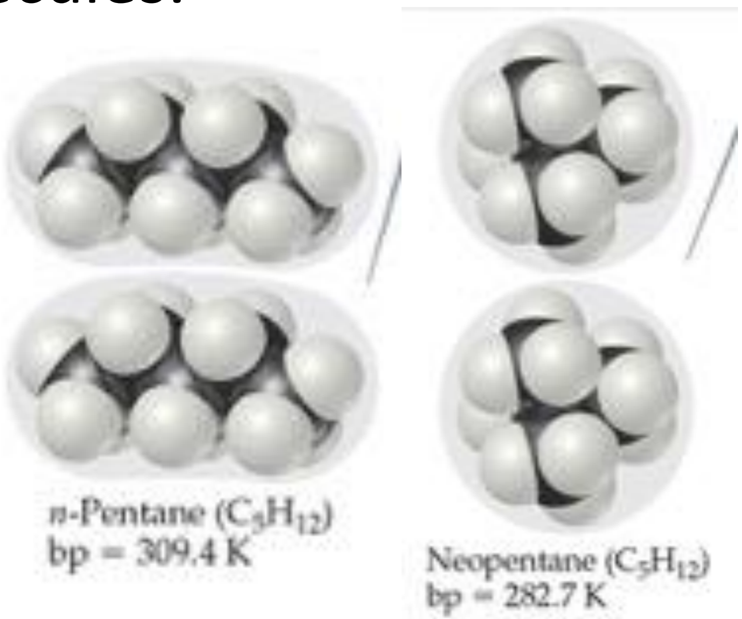
# Dispersion Forces

- **Dispersion force** strength depends on the **polarizability** of the atoms or molecules.

**Ease of distortion of electrons in the molecule or atom**

**increases with larger number and less tightly held electrons in the atom or molecule.**

**tends to increase with increasing molecular weight**




- Shape of molecules with similar masses (more compact, less dispersion forces)

| | M. wt | Melting Pt. (℃) | Boiling Pt. (℃) |
|---|---|---|---|
| He | 4 | -270 | -269 |
| Ne | 20 | -249 | -246 |
| Ar | 39.9 | -189 | -186 |
| Kr | 83.8 | -157 | -153 |
| Xe | 131.3 | -112 | -108 |
| | | | |
| $F_2$ | 38 | -220 | -188 |
| $Cl_2$ | 71 | -101 | -34 |
| $Br_2$ | 159.8 | -7 | 59 |
| $I_2$ | 153.8 | 114 | 184 |

**As molecular weight increases**

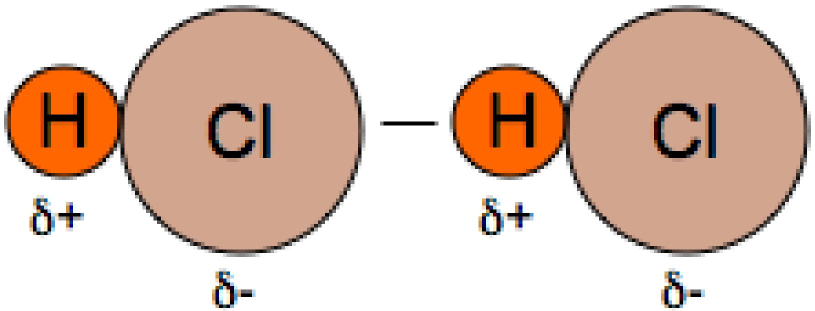
**Stronger dispersion forces**

**Higher boiling & melting points**

**Ex. List $CCl_4$, $CBr_4$, and $CH_4$ in order of increasing boiling point**

# Dipole-Dipole Forces

- **Attractive force between polar molecules.**




Electrostatic attraction between $\delta^-$ & $\delta^+$

The larger the dipole moment (higher polarity) and the closer the molecules, the greater is the force

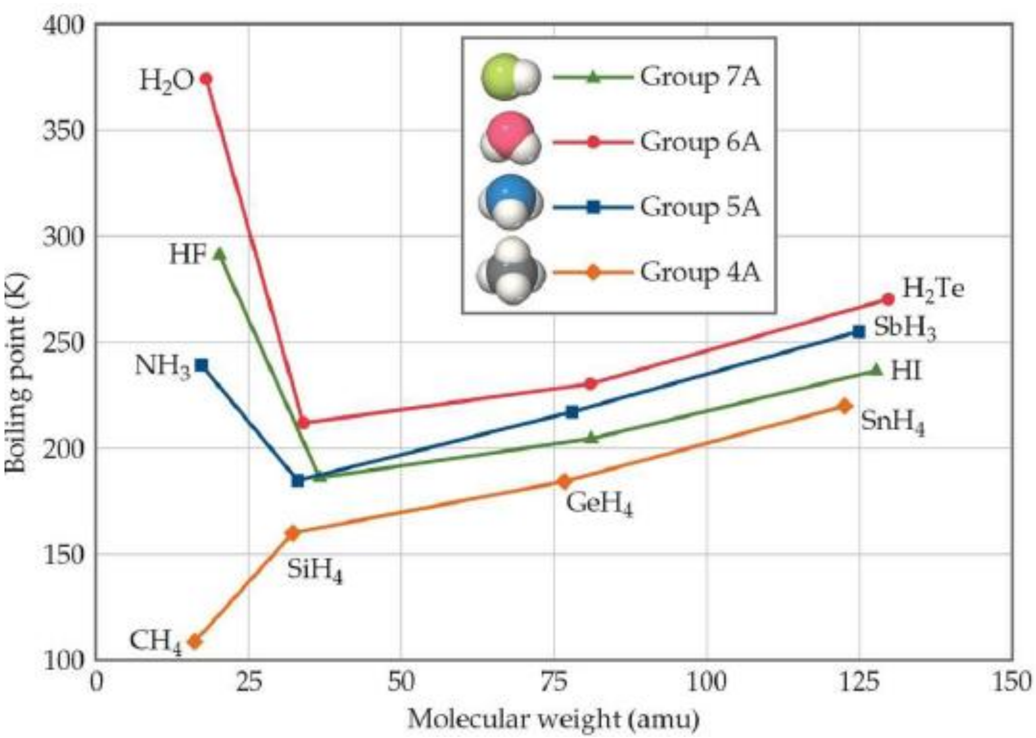
- It is stronger than London dispersion forces.
- As Dipole-Dipole force increases the boiling point increases.

| | | | |
|---|---|---|---|
| propane $CH_3CH_2CH_3$ | dimethyl ether $CH_3OCH_3$ | acetaldehyde $CH_3CHO$ | acetonitrile $CH_3CN$ |
| 44 g/mol | 46 g/mol | 44 g/mol | 41 g/mol |
| 0.1 | 1.3 | 2.7 | 3.9 |
| 231 K | 248 K | 294 K | 355 K |

Lowest boiling point K

Highest boiling point K

| $XH_4$ | Bpt /K | $XH_3$ | Bpt/ K | $H_2X$ | Bpt/ K | HX | Bpt/ K | atom | Bpt/ K |
|---|---|---|---|---|---|---|---|---|---|
| $CH_4$ | 112 | $NH_3$ | 240 | $H_2O$ | 373 | HF | 293 | Ne | 27 |
| $SiH_4$ | 161 | $PH_3$ | 185 | $H_2S$ | 212 | HCl | 188 | Ar | 87 |
| $GeH_4$ | 184 | $AsH_3$ | 218 | $H_2Se$ | 232 | HBr | 206 | Kr | 121 |
| $SnH_4$ | 221 | $SbH_3$ | 256 | $H_2Te$ | 271 | HI | 238 | Xe | 166 |




- ➢ **$NH_3$, $H_2O$** and **HF** show considerably higher boiling points than 'expected. **Why**?
- ➢ These anomalous boiling points are accounted for the phenomena of **hydrogen bonding**.
- ➢ $H_2O$ has high $C_s$, $\Delta H_{vap}$ & $T_b$.

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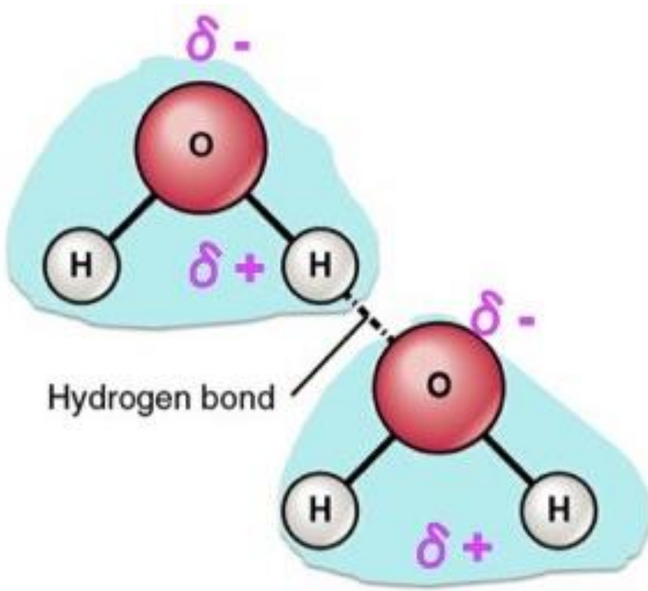
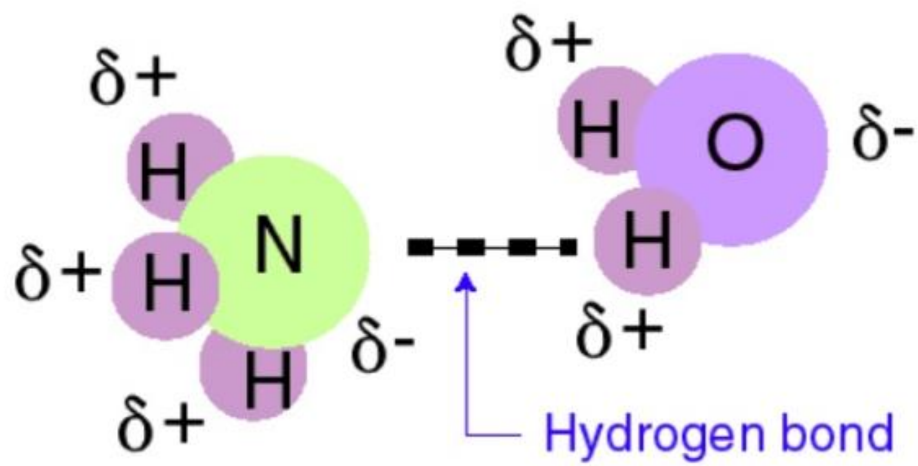
# Hydrogen Bonding

➢ A special type of dipole-dipole interaction between H-atom in an **H-X** molecule and a lone pair of electrons on the the **X** atom in another molecule.

**X = N, O, or F** **the most electronegative atoms.**

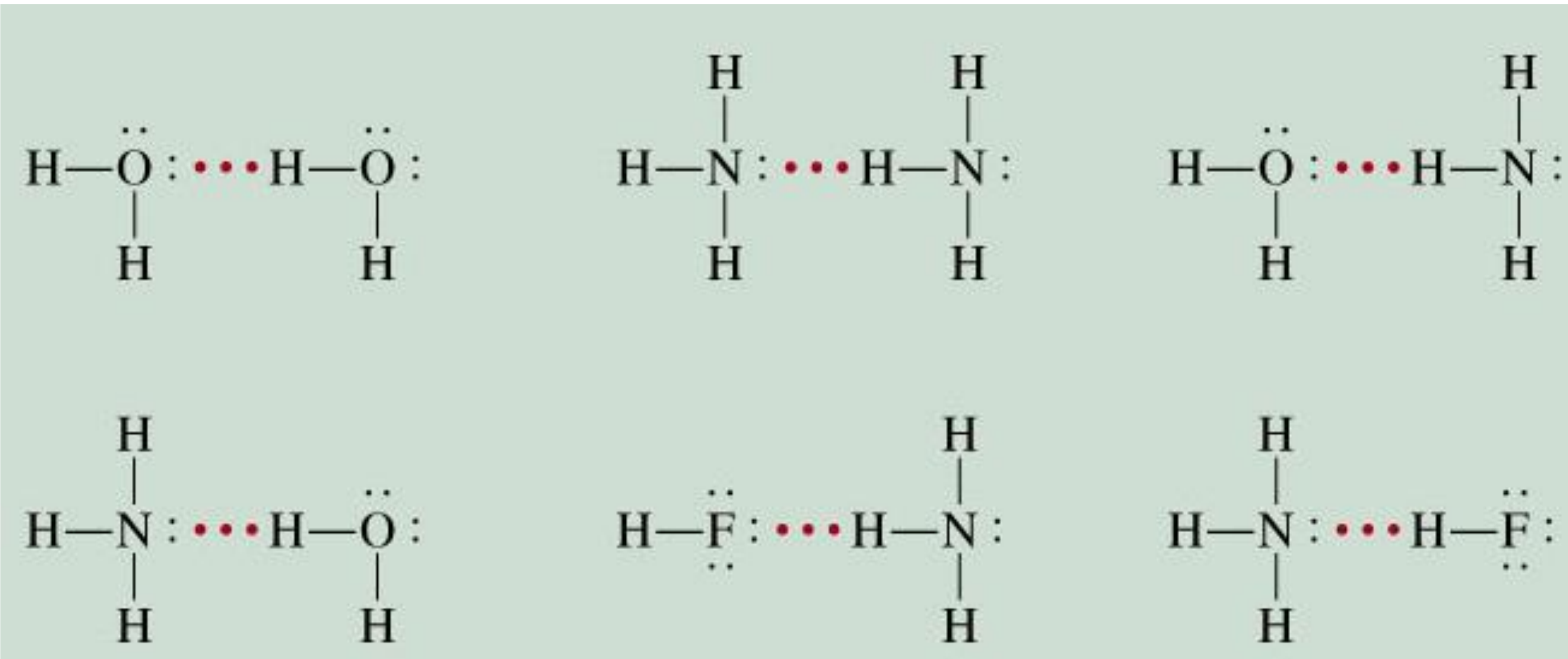



**H-Bonding between two $H_2O$ molecules**




**H-Bonding between $H_2O$ & $NH_3$ molecules**

# Hydrogen Bonding



# Hydrogen Bonding

- Strength determined by coulombic interaction between the lone-pair of electrons of the electronegative atom and the H-nucleus on the other atom.

- Stronger than normal dipole-dipole interaction and weaker than ordinary chemical bond.

| Bond | Bond Energy (kcal/mol) |
|---|---|
| Covalent | 50 - 150 |
| Ionic | 5 - 10 |
| Hydrogen | 2 - 5 |
| Van der Waals | 0.5 |

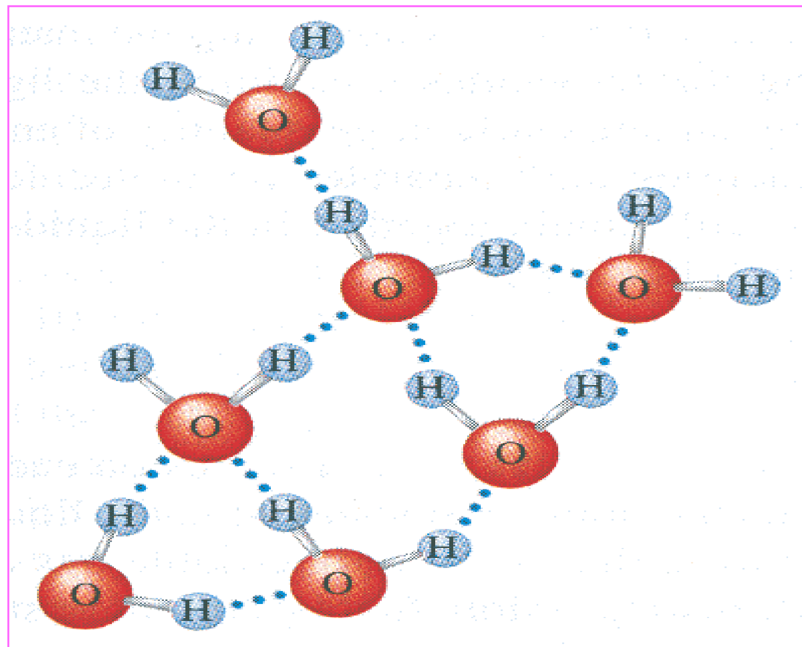
# Hydrogen Bonding

- Strength increases with increasing electronegativity and decreasing size of **X** atom.

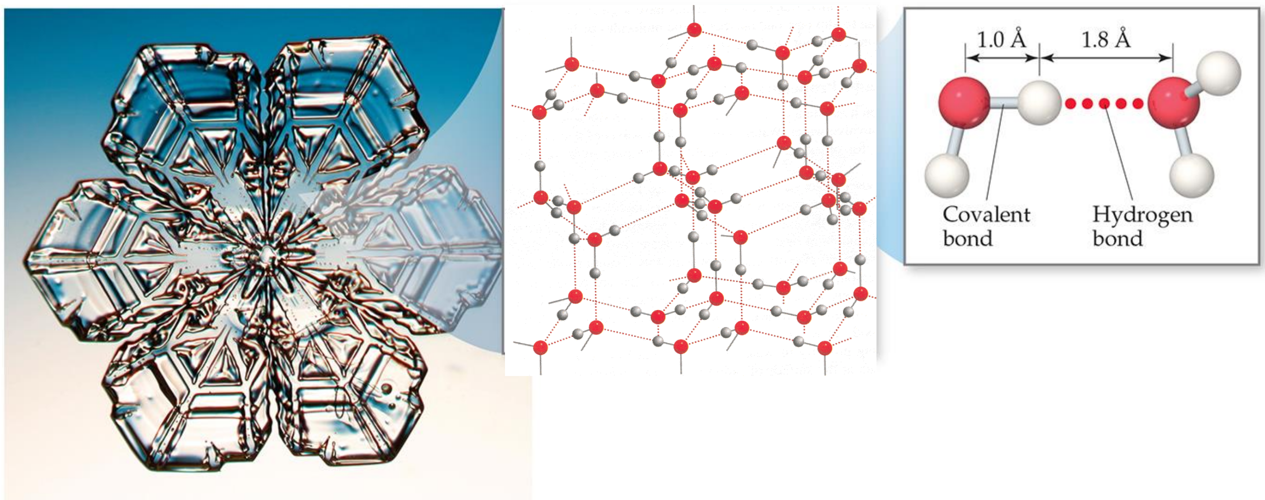
**Expect:** **HF > $H_2O$ > $NH_3$ ?**


**Actually:** **$H_2O$ > HF > $NH_3$**

Highest in $H_2O$ because each O-atom has two pairs of unshared electrons and forms two H-bonds with two other molecules forming 3-dimension network.



# Ice Compared to Liquid Water

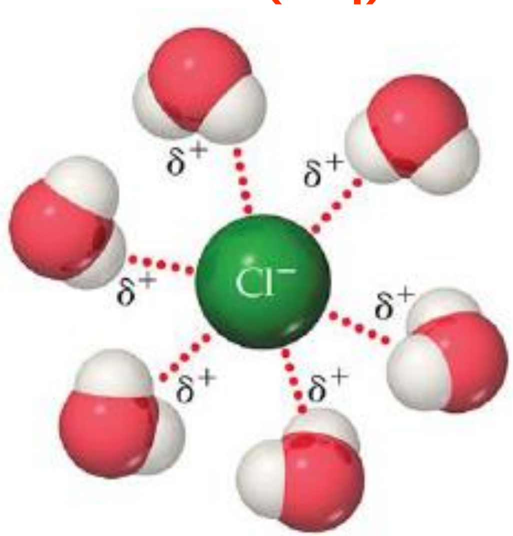


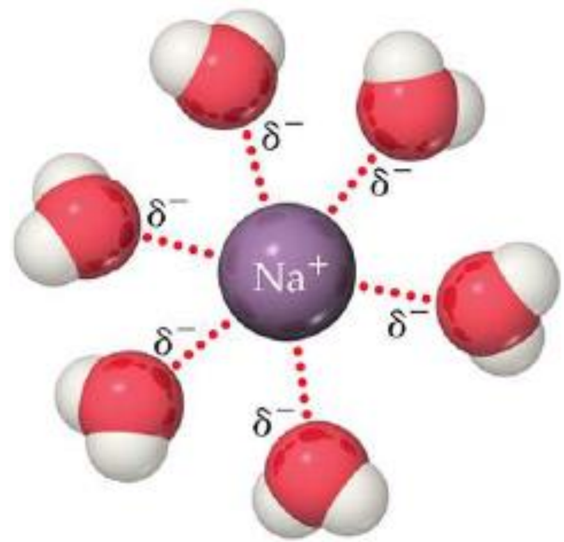

- Solid Water Less Dense Than Liquid Water
- The interactions in the liquid are random. However, when water freezes, the molecule assume the ordered.
- Each O-atom forms two H-bonds, forming a three-dimensional network with large empty cavities.

# Ion-Dipole Forces

- Attractive forces between an <u>ion</u> (+ or -) and a <u>polar</u> molecule
- Found in solutions of ionic compounds

$$NaCl(aq) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$




Positive ends of polar molecules are oriented towards negatively charged anion

Negative ends of polar molecules are oriented towards positively charged anion

# Determining Intermolecular Forces in a Substance

| Type of intermolecular interaction | Atoms Examples: Ne, Ar | Nonpolar molecules Examples: $BF_3$, $CH_4$ | Polar molecules without OH, NH, or HF groups Examples: HCl, $CH_3CN$ | Polar molecules containing OH, NH, or HF groups Examples: $H_2O$, $NH_3$ | Ionic solids dissolved in polar liquids Examples: NaCl in $H_2O$ |
|---|---|---|---|---|---|
| Dispersion forces (0.1–30 kJ/mol) | √ | √ | √ | √ | √ |
| Dipole–dipole interactions (2–15 kJ/mol) | | | √ | √ | |
| Hydrogen bonding (10–40 kJ/mol) | | | | √ | |
| Ion–dipole interactions (>50 kJ/mol) | | | | | √ |

- Note that ALL chemicals exhibit dispersion forces.

**Exercise:** List the substances $BaCl_2$, $H_2$, CO, HF and Ne in order of increasing boiling points.

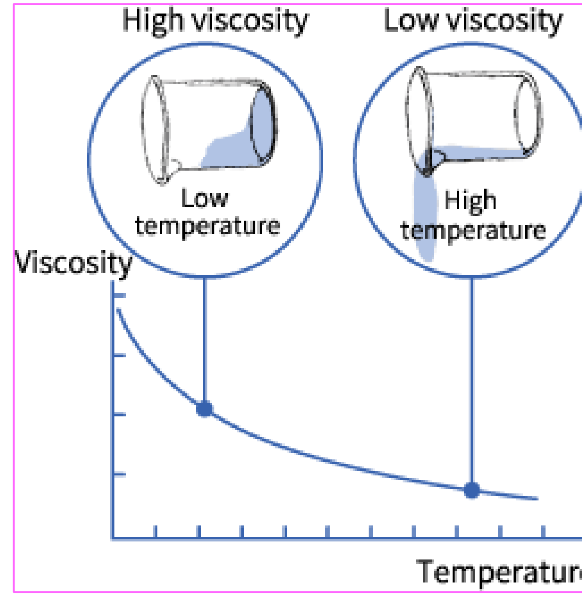
**What type(s) of intermolecular forces exist between each of the following molecules?**

| | | | |
|---|---|---|---|
| HBr | Polar | Dipole-dipole forces<br>Dispersion forces | dominant forces |
| $CH_4$ | Nonpolar | Dispersion forces | dominant forces |
| HF | Polar | H-bonding<br>Dipole-Dipole forces<br>Dispersion forces | dominant forces |
| $CH_3F$ | Polar | Dipole-Dipole forces<br>Dispersion forces | dominant forces |
| $NH_2NH_2$ | Polar | H-bonding<br>Dipole-Dipole forces<br>Dispersion forces | dominant forces |

# 11.3 Properties of Liquids

## 1. Viscosity

- **Resistance of a liquid to flow.**


- Increases with increasing IMF in the liquid and decreases with increasing temperature.

<u>Exercise:</u> List the substances $C_6H_6$, $CCl_4$, $H_2O$, $HOCH_2CH(OH)CH_2OH$ in order of increasing viscosity.

$C_6H_6 = 6.25 \times 10^{-4}$

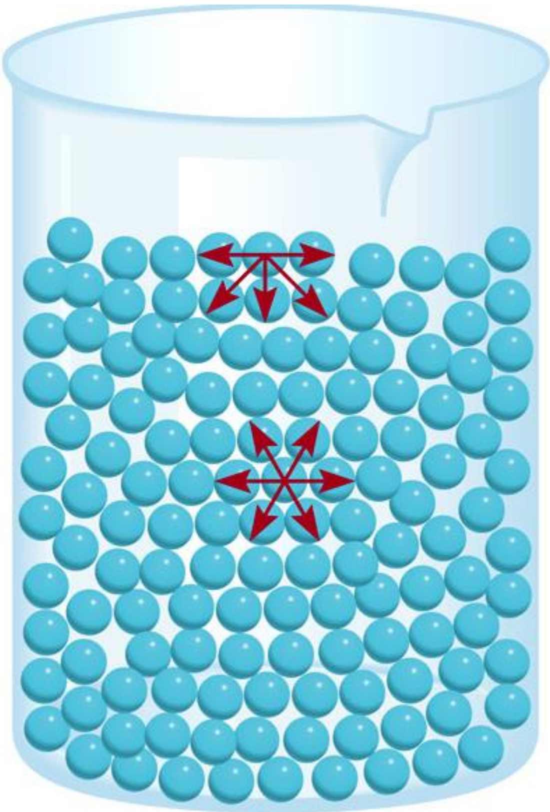
$CCl_4 = 9.69 \times 10^{-4}$

$H_2O = 1.01 \times 10^{-3}$

$HOCH_2CH(OH)CH_2OH = 1.49$

# 2. Surface Tension

- Results from imbalance of IMF at the surface of the liquid.
- Increases with increasing IMF in the liquid.
- $H_2O$ has higher surface tension than most liquids due to the high IMF (H-bonding) between molecules.

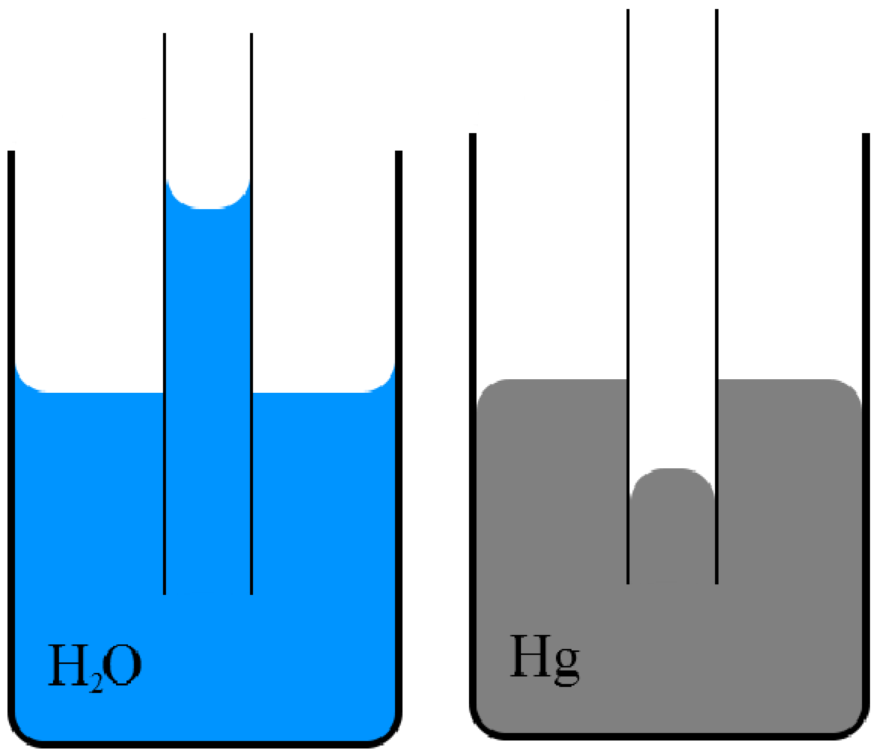


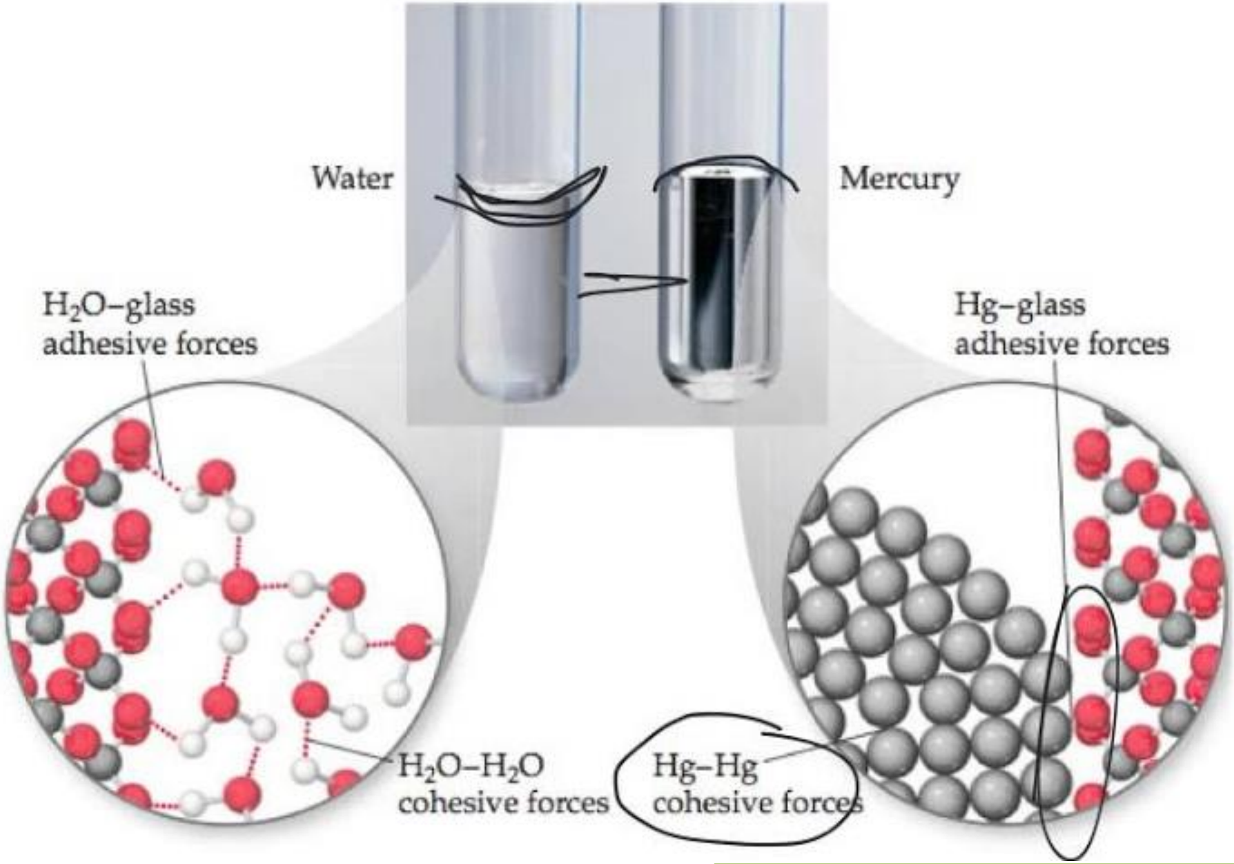
# Examples of Surface Tension

**Capillary Action**: the rise of liquids up very narrow tubes.

**Cohesive Forces:** intermolecular attractions between similar molecules.

**Adhesive Forces:** attraction between unlike molecules.






Because adhesive > Cohesive, $H_2O$ molecules touching glass adhere to the wall more than to each other, therefore forming convex surface

Because Cohesive > adhesive, Hg molecules touching glass adhere to the wall less than to each other, therefore forming concave surface

# 11.4 Phase Changes (Change of State)

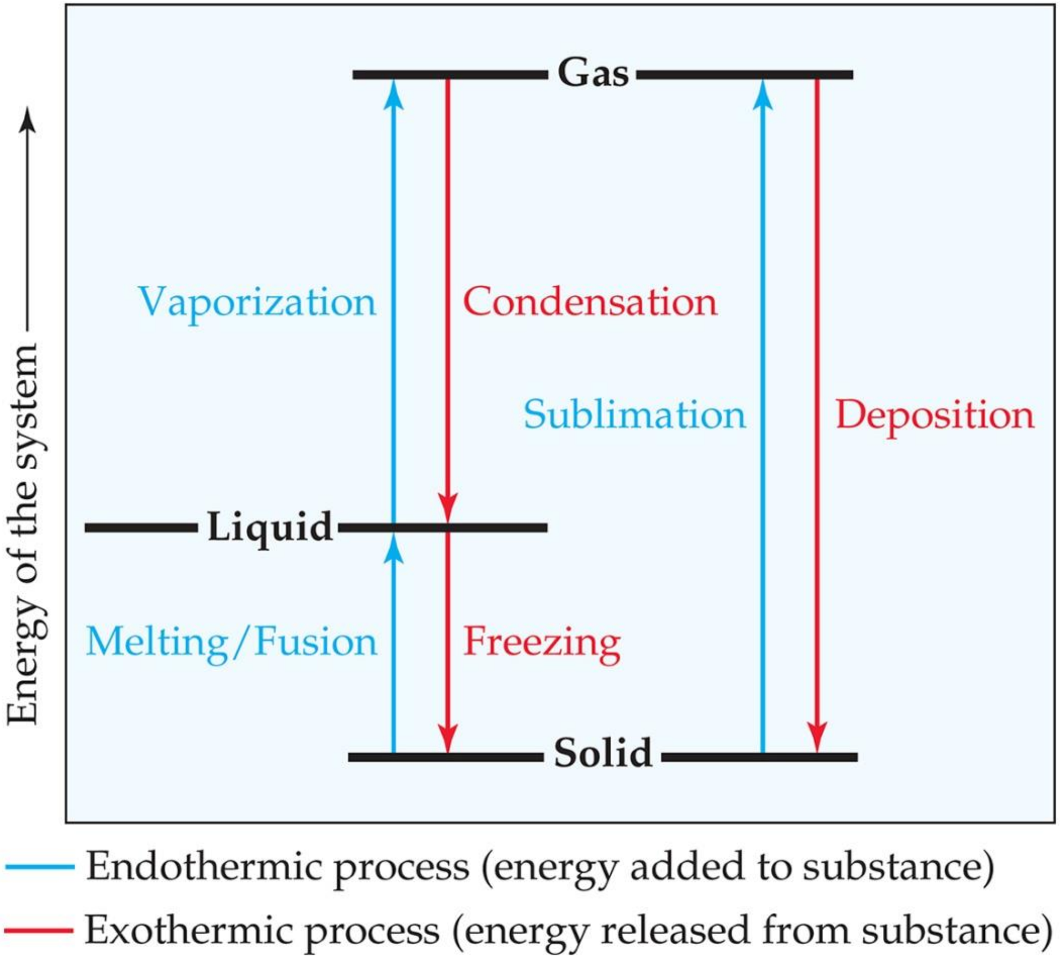
- Conversion from one state of matter to another.
- Examples of Phase Changes:

  Melting $\rightleftarrows$ freezing

  Vaporizing $\rightleftarrows$ condensing

  Subliming $\rightleftarrows$ depositing

- Energy is either added or released in a phase change.

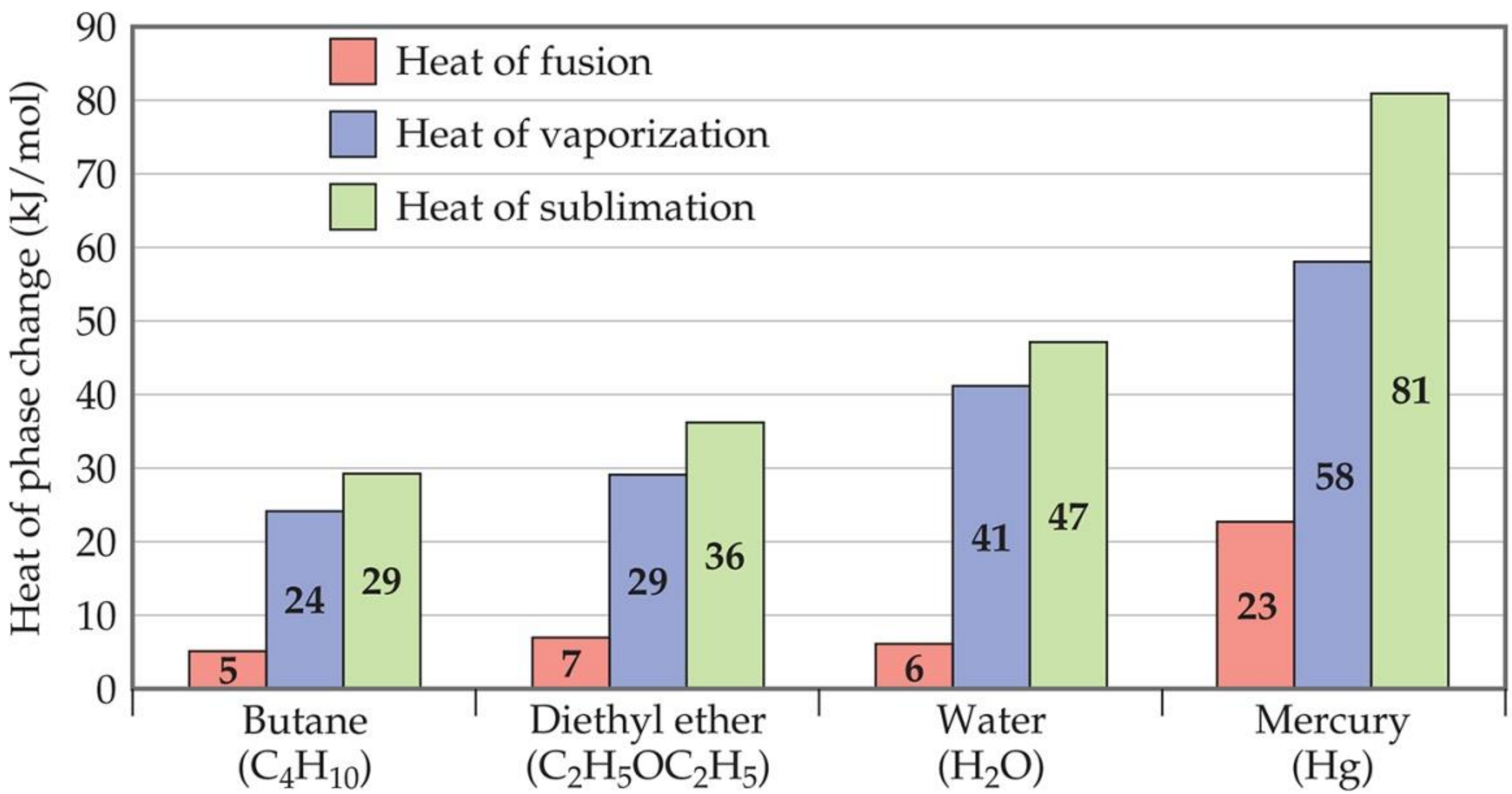



**Heat of fusion (Enthalpy of Fusion), $\Delta H_{fus}$:** energy required to change a solid at its melting point to a liquid.

**Heat (Enthalpy) of vaporization, $\Delta H_{vap}$:** energy required to change a liquid at its boiling point to a gas.

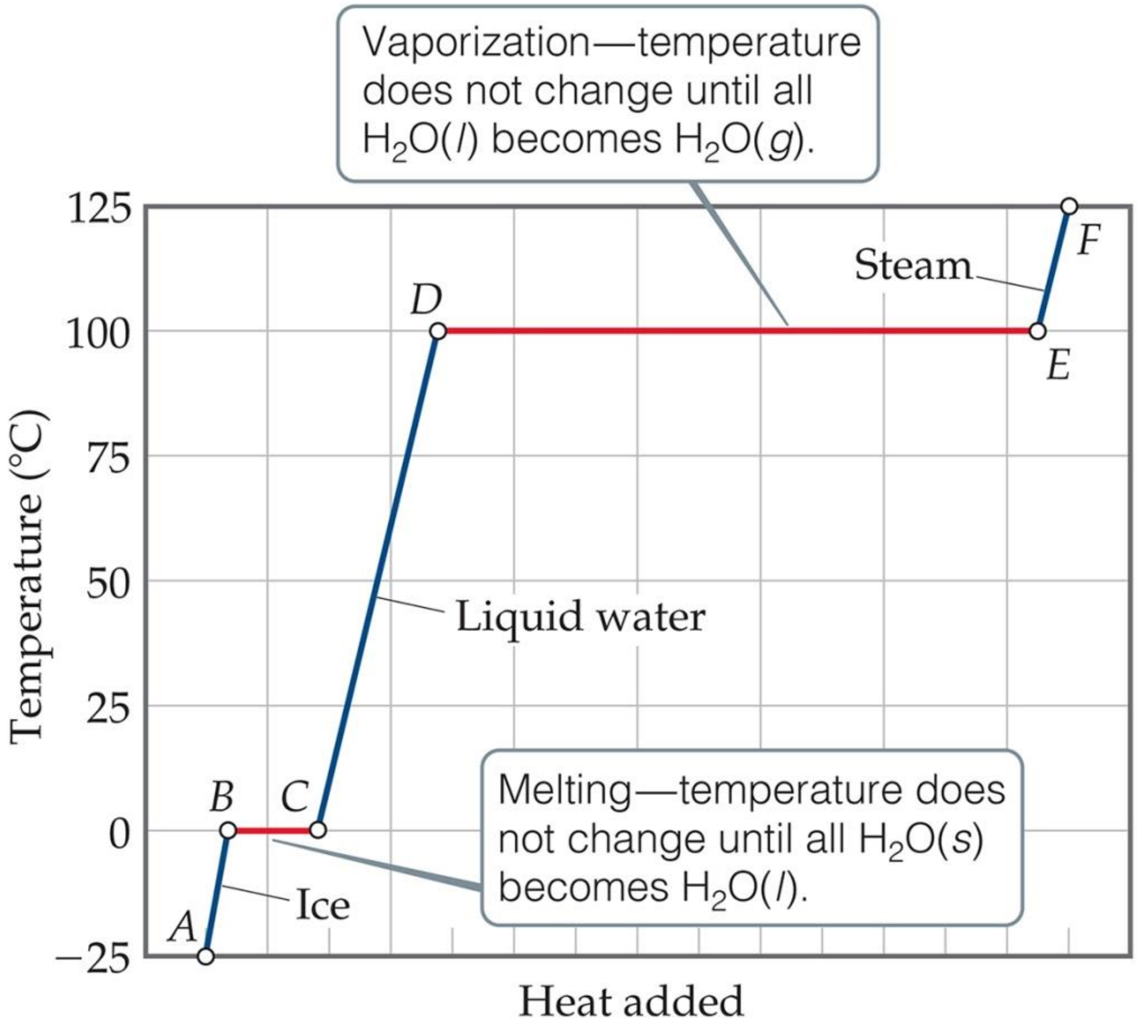
**Heat (Enthalpy) of sublimation, $\Delta H_{sub}$:** energy required to change a solid directly to a gas.

# Energy Change and Change of State




**Notice that the $\Delta H_{fusion}$ < $\Delta H_{vaporization:}$**: less energy is required to allow particles to move past one another than to separate them totally.

# Heating Curves

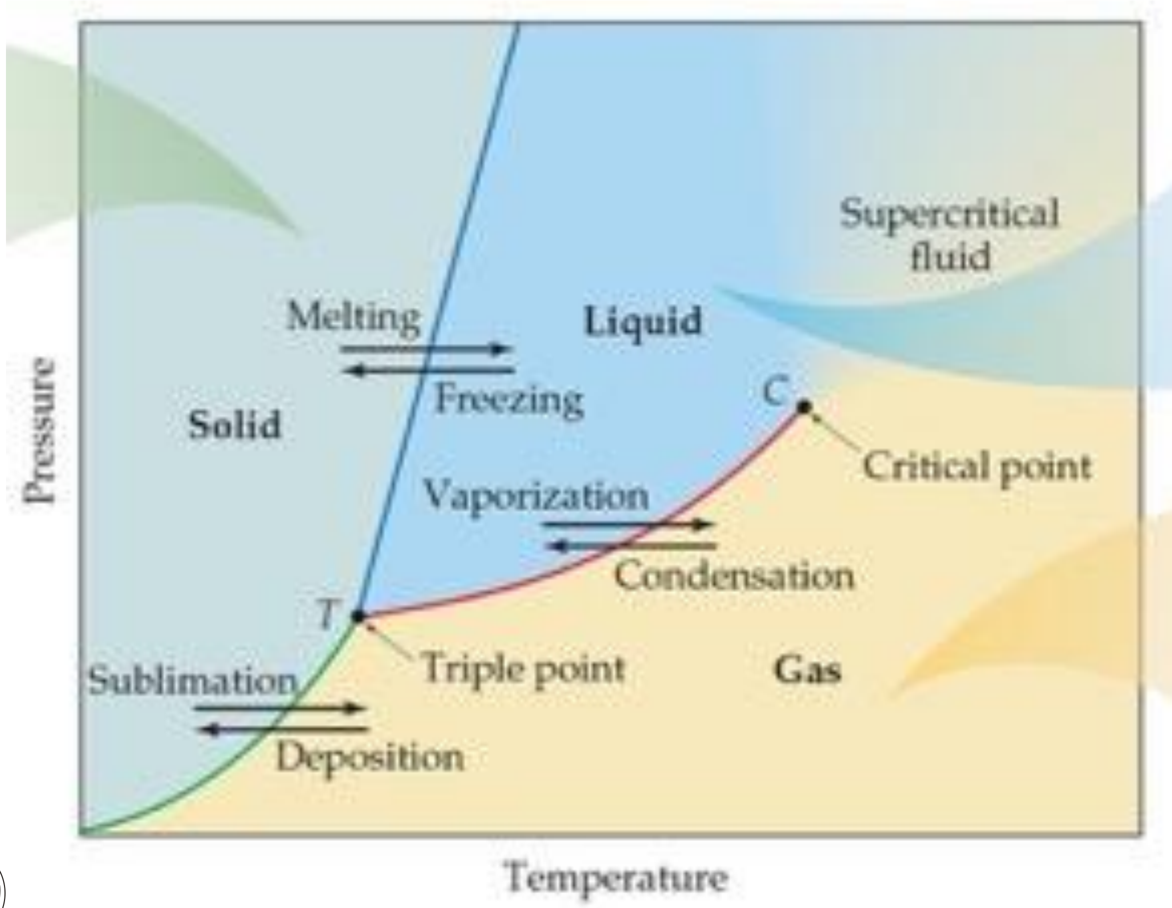



- A graph of temperature vs. heat added is called a **heating curve.**
- The temperature of the substance does not rise during a phase change.
- Within a phase:

**Heat (q) = $C_s$ . m . $\Delta T$**

# Supercritical Fluids

- When the temperature exceeds the $T_c$ and the pressure exceeds $P_c$ the liquid and gas phases are indistinguishable from each other, and the substance is in a state called a **supercritical fluid**.




supercritical fluids can behave as solvents

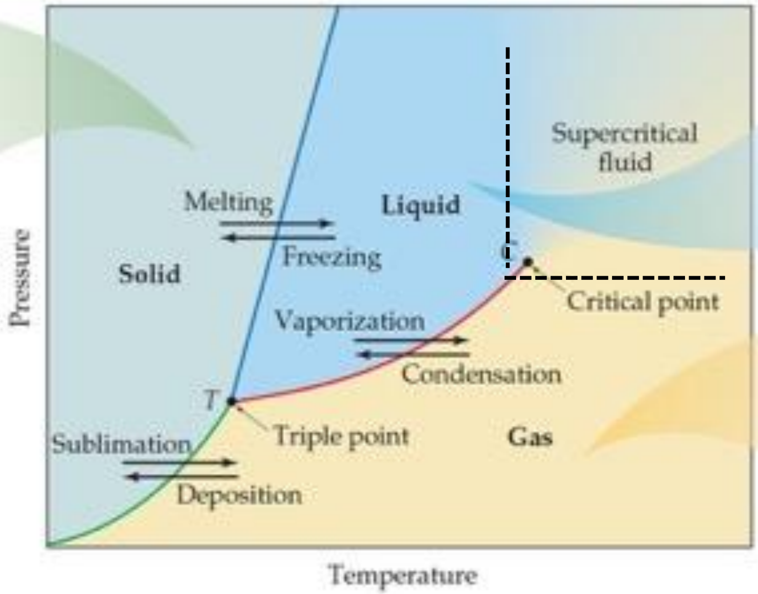
# Critical Temperature and Pressure

- **Critical temperature ($T_c$)**: the highest temperature at which a distinct liquid phase can form.
- **Critical pressure ($P_c$)**: the pressure required to liquify a gas at its critical temperature.

The greater the IMF, the higher the $T_c$ & $P_c$ of a substance.

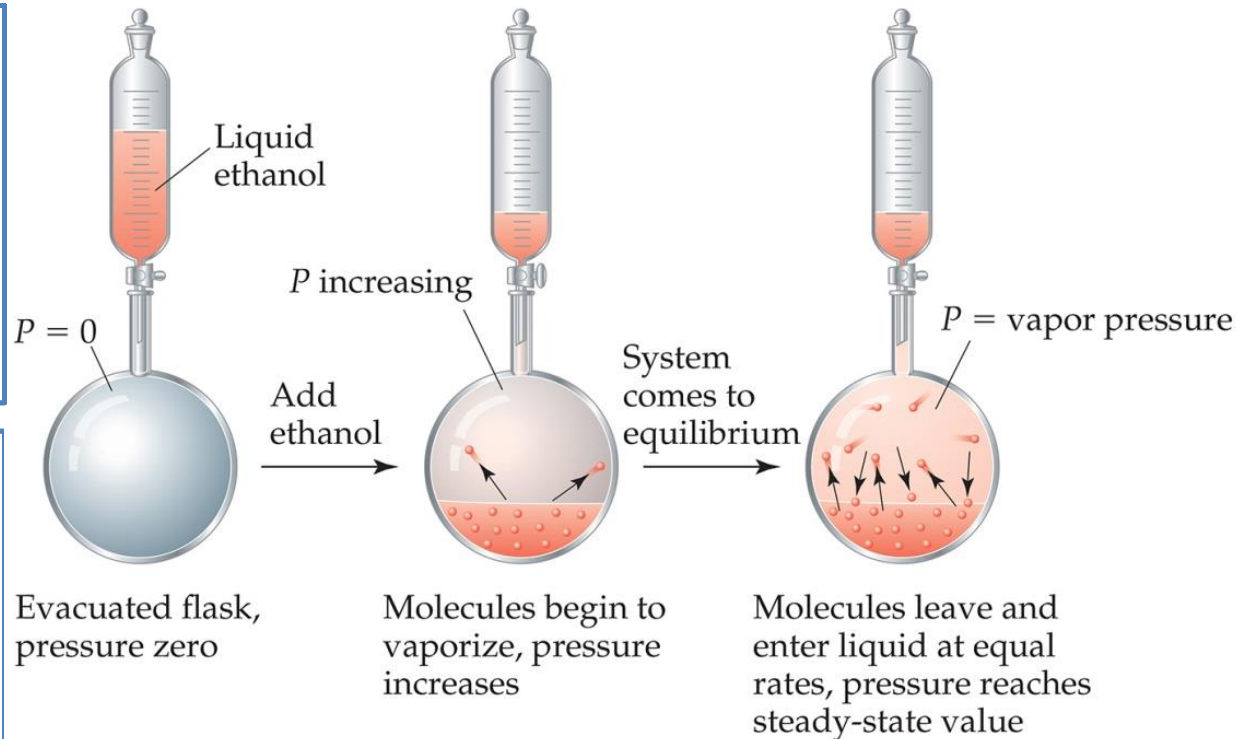
**TABLE 11.7** Critical Temperatures and Critical Pressures of Selected Substances

| Substance | $T_c$ (°C) | $P_c$ (atm) |
|---|---|---|
| Ammonia ($NH_3$) | 132.4 | 111.5 |
| Argon (Ar) | −186 | 6.3 |
| Benzene ($C_6H_6$) | 288.9 | 47.9 |
| Carbon dioxide ($CO_2$) | 31.0 | 73.0 |
| Ethanol ($C_2H_5OH$) | 243 | 63.0 |
| Diethyl ether ($C_2H_5OC_2H_5$) | 192.6 | 35.6 |
| Mercury (Hg) | 1462 | 1036 |
| Methane ($CH_4$) | −83.0 | 45.6 |
| Molecular hydrogen ($H_2$) | −239.9 | 12.8 |
| Molecular nitrogen ($N_2$) | −147.1 | 33.5 |
| Molecular oxygen ($O_2$) | −118.8 | 49.7 |
| Sulfur hexafluoride ($SF_6$) | 45.5 | 37.6 |
| Water ($H_2O$) | 374.4 | 219.5 |



# 11.5 Vapor Pressure

- As more molecules escape from the liquid, the pressure they exert increases.

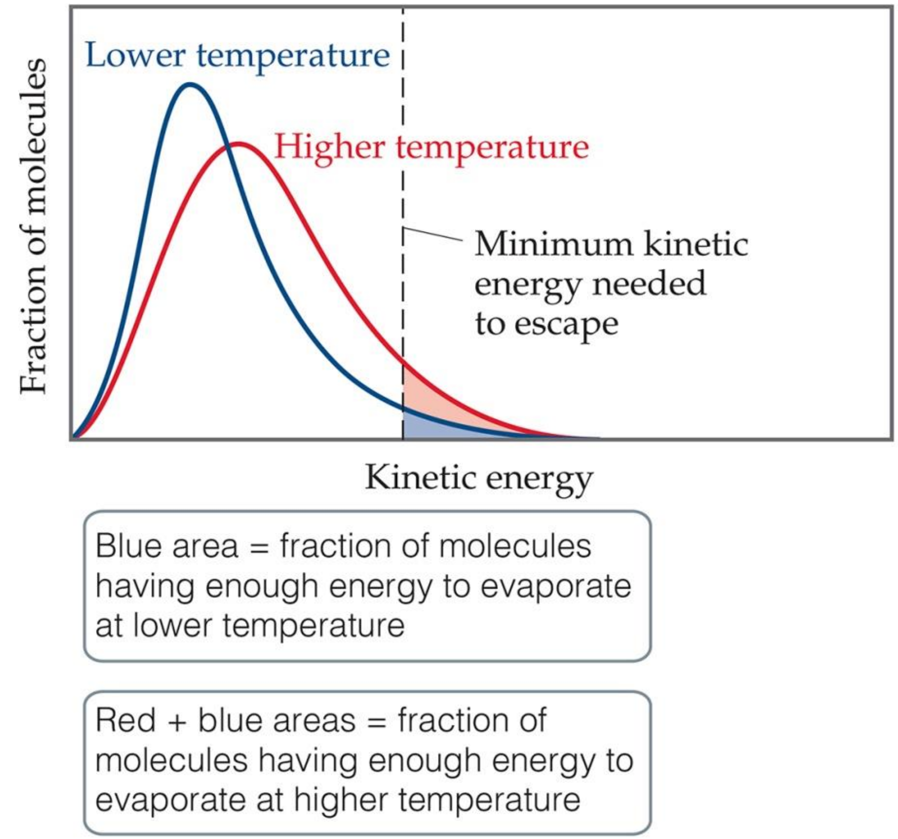



- The liquid and vapor reach a state of **dynamic equilibrium:** liquid molecules evaporate and vapor molecules condense *at the same rate*.

- The **vapor pressure** of a liquid is the pressure exerted by its vapor when the liquid and vapor are in a dynamic equilibrium.

# 11.5 Vapor Pressure

- At any temperature, some liquid molecules have enough energy to escape the surface and become a gas.
- Liquids that evaporate readily are said to be volatile.
- As the temperature rises, the fraction of molecules that have enough energy to break free increases.




- The vapor pressure increases with increasing temperature.

# Vapor pressure and boiling point

- The **boiling point** of a liquid is the temperature at which its vapor pressure equals external pressure acting on the liquid surface.
- The **normal boiling** point of a liquid is the temperature at which its vapor pressure is 760 torr.